

Solution Requirements

Date	26 June 2026
Team ID	
Project Name	Machine Learning-Based Flood Prediction and Early Warning System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements

This document defines the **Functional Requirements (FR)** and **Non-Functional Requirements (NFR)** for the Machine Learning-Based Flood Prediction and Early Warning System.

Step 1: Functional Requirements (FR)

S.NO	Requirement Category	Requirement Category	Priority
1	Weather Data Input	Accept weather details from the user.	High
2	Data Preprocessing	Clean and prepare input data.	High
3	Model Integration	Connect Flask with the trained ML model.	High
4	Flood Prediction	Predict whether flooding is likely.	High
5	Result Display	Show prediction on the web page.	Medium
6	Model Evaluation	Compare ML models and select the best one.	High
7	Early Warning	Support timely flood alerts.	High
8	Prediction Logs	Store prediction history.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric/ Acceptance Criteria
1	Performance & Speed	Fast prediction.	< 2 seconds
2	Scalability	Support high applicant volume.	1000+ requests
3	Security & Privacy	Encrypt sensitive data.	Secure Access
4	Reliability & Availability	High availability.	99.9% uptime
5	Usability & Accessibility	Responsive and easy UI	User friendly
6	Other	Maintainability and portability.	Modular design